

Write in EEPROM

```
BCF    EECON1, EEPGD      :: for location move
BCF    EECON1, CFGS       value in EEADR
BSF    EECON1, WREN       :: for data move
BCF    INTCON, GIE        value in EEDATA
MOVLW  0X55
MOVWF  EECON2
BSF    EECON1, WR
BSF    INTCON, GIE
BCF    EECON1, WREN
```

Read in EEPROM

```
BCF    EECON1, EEPGD      :: for location move
BCF    EECON1, CFGS       value in EEADR
BSF    EECON1, RD
NOP
MOVFF  EEDATA, PORTB
```

→ BTFSS INTCON, TMR0IF → Timer 0 interrupt check
Retfie

→ BTFSS PIR1, TMR1IF → Timer 1 interrupt check.

→ BTFSC PIR1, TXIF → Interrupt due to TxD.

Interrupt	Flag bit	Register	Enable bit	Register
Timer 0	TMR0IF	INTCON	TMR0IE	INTCON
Timer 1	TMR1IF	PIR1	TMR1IE	PIE1
Timer 2	TMR2IF	PIR1	TMR2IE	PIE1
Timer 3	TMR3IF	PIR3	TMR3IE	PIE2

INT 0 (RB0)	INT0IF	INTCON	INT0IE	INTCON
INT 1 (RB1)	INT1IF	INTCON3	INT1IE	INTCON3
INT 2 (RB2)	INT2IF	INTCON3	INT2IE	INTCON3

PortB change interrupt

↳ RB4-RB7 → single flag (RBIF)

High priority → 0008

Low priority → 0018

To make low priority of timer interrupt

BCF IPR1, TMR1IE
BSF RCON, IPEN
BSF INTCON, GIEH
BSF INTCON, GIEH

∴ IPEN=0 interrupt are directed to default location
∴ IPEN=1 set low and high priority

Main

Timer 0

CLRF TRISD

SETF TRISC

MOVLW 0x8

MOVWF TOCON

high nibble MOVWF 0xFF

low MOVWF TMR0H

nibble MOVLW 0xF2

values MOVWF TMR0L

BCF INTCON, TMR0IF

→

BSF TOCON, TMR0ON

BSF INTCON, TMR0IE

BSF INTCON, GIE

→ 6 lines left in Main program.
Interrupt code

e.g.

OVER MOVFF PORTC, PORTD
BRA OVER

Timer 1

MOVWF 0x00

MOVWF T1CON

MOVLW 0xFF

MOVWF TMR1H

MOVLW 0xF2

MOVWF TMR1L

BCF PIR1, TMR1IF

→

BSF PIE1, TMR1IE

~~BSF PIE1,~~

BSF INTCON, PEIE

BSF T1CON, TMR1ON

→ Jo kam Main may
karna ho us ka code
likh lena.

ISR

Load values of higher
and lower nibble

Task karna hai

BCF INTCON, TMR0IF } always
GOTO CHK-INT } write.

Same as TMR0

Task karna hai

BCF PIR1, TMR0IF

GOTO CHK-INT

- ADDLW
- ANDLW
- DAW
- SUBLW
- XORLW
- MULLW
- ADDWF fileReg, d
- ADDWFC fileReg, d
- ANDWF fileReg, d
- IORWF fileReg, d
- SUBFWB fileReg, d
- SUBWF fileReg, d
- SUBWFB fileReg, d
- XORWF fileReg, d
- COMF fileReg, d
- DECF fileReg, d
- DECFSZ fileReg, d
- DECFSNZ fileReg, d
- INCF fileReg, d
- INCFSZ fileReg, d
- INCSNZ fileReg, d
- MOVF fileReg, d
- NEGF fileReg, d
- RLCF fileReg, d
- RLNCF fileReg, d
- RRCF fileReg, d
- RRNCF fileReg, d
- SWAPF fileReg, d
- BTG fileReg, d
- IORLW
- MOVFF
- BC
- BNC
- BZ
- BNZ
- BN
- BNC
- BOV
- BNOV
- BRA
- GOTO
- CALL
- RCALL
- RETURN
- NOP
- CLRF
- SETF
- BCF fileReg, bit
- BSF fileReg, bit

- BTFSS fileReg, bit
- BTFSC fileReg, bit
- BTG fileReg, bit
- STATUS
- C - Carry flag
- DC - Digital Carry flag
- Z - Zero flag
- OV - Overflow flag
- N - Negative flag
- CPFSGT
- CPFSEQ
- CPFSLT
- EQU
- ORG

- TMR0H/TMR1H
- TMR0L/TMR1L
- TMR0IF/TMR1IF
- INTCON
- PIR1

T0CON:-

- TMR0ON D7 Timer0 ON and OFF control bit
- T08BIT D6 Timer0 8-bit/16-bit selector bit
- T0CS D5 Timer0 clock source select bit
- T0SE D4 Timer0 source edge select bit
- PSA D3 Timer0 prescaler assignment bit
- T0PS2:T0PS0 D2D1D0 Timer0 prescaler selector

T1CON:-

- RD16 D7 16-bit read/write enable bit
- TICKPS2:T1CKPS0 D5D4 Timer1 prescaler selector
- T1OSCEN D3 Timer1 oscillator enable bit
- T1SYNC D2 Timer1 synchronization
- TMRICS D1 Timer1 clock source select bit
- TMR1ON D0 Timer1 ON and OFF control bit